

NEW SEMIPARAMETRIC AND NONPARAMETRIC BOOTSTRAP TESTS FOR SPEARMAN’S RHO

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Summary: In this paper we propose two bootstrap-based tests for Spearman’s rho. The first is a semiparametric test based on copulas, while the second is a nonparametric test. The efficiency of the tests are investigated by means of a Monte-Carlo study. It is found that they perform very satisfactorily as far as size and power are concerned.

1. Introduction

Hall and Wilson (1991) highlighted some important guidelines for bootstrap hypothesis testing. Their first guideline states that, when one wants to estimate the critical value, resampling must be done in a way that reflects the null hypothesis. This must be done even if the data were generated from a distribution specified by the alternative hypothesis. This guideline is crucial to the success of bootstrap hypothesis testing and have been mentioned by, among others, Beran (1988), Hinkley (1988), Fisher and Hall (1990) and Westfall and Young (1993). For some tests, however, it is not so easy to “mimic” the null

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hypothesis when resampling, and careful thought should be given as to how this resampling might take place.

In this paper we propose two bootstrap-based tests for Spearman's rho. The first test is a semiparametric test based on copulas, while the second is a nonparametric test. The content of the article is arranged in the following way: Section 2 introduces some basic concepts. In Sections 3 and 4 we propose new semiparametric and nonparametric bootstrap tests for Spearman's rho, respectively. The paper concludes with the results of a Monte-Carlo study and some final remarks.

2. Basic concepts

The nonparametric measure of association known as Spearman's rho was first discussed by Spearman (1904). If X and Y are random variables with respective distribution functions $F(x)$ and $G(y)$, then the population version of Spearman's rho (denoted by ρ_s) is defined to be the Pearson correlation coefficient (denoted by ρ_p) between the random variables $F(X)$ and $G(Y)$:

$$\rho_s(X, Y) = \rho_p(F(X), G(Y)). \quad (1)$$

Because of this fact, Spearman's rho is sometimes referred to as the rank or grade coefficient of correlation and has the advantage that it is not influenced by outliers. Furthermore, the population version of Spearman's rho can be expressed in terms of a copula C associated with the vector (X, Y) as follows (Schweizer and Wolff, 1981):

$$\begin{aligned} \rho_s(X, Y) &= 12 \int_0^1 \int_0^1 C(u, v) du dv - 3 \\ &= 12 \int_0^1 \int_0^1 uv dC(u, v) - 3. \end{aligned} \quad (2)$$

Now, let $\mathbf{Z}_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ be a random sample from a population (X, Y) and denote by $H(x, y)$ the joint distribution function of (X, Y) . As above, let $F(x)$ and $G(y)$ be the marginal distribution functions of X and Y , respectively.

Consider, without loss of generality, the right-sided alternative hypothesis

$$H_0 : \rho_s(X, Y) = \rho_{s,0} \quad \text{vs.} \quad H_A : \rho_s(X, Y) > \rho_{s,0}.$$

Suppose the test rejects H_0 if and only if

$$T_n(\mathbf{Z}_n) \geq C_n(\alpha),$$

where $C_n(\alpha)$ is a critical value defined as

$$P_{H_0}(T_n(\mathbf{Z}_n) \geq C_n(\alpha)) \cong \alpha,$$

with $T_n(\mathbf{Z}_n)$ an appropriate test statistic, and $\alpha(0 < \alpha < 1)$ the prescribed nominal significance level of the test.

Possible test statistics for this testing scenario are

$$T_n^{(1)}(\mathbf{Z}_n) = \hat{\rho}_s(\mathbf{Z}_n) \quad \text{and} \quad T_n^{(2)}(\mathbf{Z}_n) = \sqrt{n-3} \left(z_{\hat{\rho}_s} - z_0 - \frac{\rho_{s,0}}{2(n-1)} \right),$$

where $\hat{\rho}_s \equiv \hat{\rho}_s(\mathbf{Z}_n)$ is an estimator for ρ_s and $z_{\hat{\rho}_s}$ and z_0 denote Fisher's Z transformation (see Fisher, 1921) given by:

$$z_{\hat{\rho}_s} = \frac{1}{2} \ln \left(\frac{1 + \hat{\rho}_s(\mathbf{Z}_n)}{1 - \hat{\rho}_s(\mathbf{Z}_n)} \right) \quad \text{and} \quad z_0 = \frac{1}{2} \ln \left(\frac{1 + \rho_{s,0}}{1 - \rho_{s,0}} \right). \quad (3)$$

If $H(x, y)$ is the bivariate normal distribution function, then $T_n^{(2)}(\mathbf{Z}_n)$ has an approximate (for large n) standard normal distribution under H_0 (Fisher, 1970) and $C_n(\alpha) \approx \Phi^{-1}(1 - \alpha)$, with Φ the standard normal distribution function. However, if $H(x, y)$ is not the bivariate normal distribution function, then the asymptotic null distribution of $T_n^{(2)}(\mathbf{Z}_n)$ is unknown (hence, also $C_n(\alpha)$).

Next, we propose a semiparametric bootstrap test for estimating $C_n(\alpha)$.

3. Semiparametric bootstrap test

Let $C_\theta(u, v)$ be a parametric family of copulas indexed by a single parameter θ (in our discussion we will only focus on single parameter families of copulas). Assume further that $F(x)$ and $G(y)$ are unknown nuisance parameters and that

$$H(x, y) = C_\theta(F(x), G(y)).$$

First, note that testing the hypothesis

$$H_0 : \rho_s(X, Y) = \rho_{s,0} \quad \text{vs.} \quad H_A : \rho_s(X, Y) > \rho_{s,0}$$

is equivalent to testing (by using the equality in (2))

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_A : \theta > \theta_0,$$

for some $\theta_0 \equiv \theta_0(\rho_{s,0})$.

The test rejects H_0 if and only if

$$\hat{\rho}_s(\mathbf{Z}_n) \geq C_n(\alpha),$$

where

$$P_{H_0}(\hat{\rho}_s(\mathbf{Z}_n) \geq C_n(\alpha)) \cong \alpha.$$

The critical value $C_n(\alpha)$ is unknown, since H is unknown (F and G are unknown). However, a bootstrap estimator $C_n(\alpha; \mathbf{Z}_n)$ can be derived for $C_n(\alpha)$ as follows:

- (1) Simulate (U_i, V_i) , $i = 1, 2, \dots, n$ from C_{θ_0} . Various methods of sampling from a given copula exist in the literature, see, e.g., Nelsen (2006). A well-known method is the so-called distribution function method:
 - (a) Generate a continuous uniform variable U from $(0, 1)$.
 - (b) Simulate V from $C_u(v) = \frac{\partial}{\partial u} C_{\theta_0}(u, v)$ as follows:
 - generate a second continuous uniform variate U_1 from $(0, 1)$, independent of U ;
 - set $V = C_U^{-1}(U_1)$.
 - (c) The pair of variables is then (U, V) .
- (2) Estimate $F(x)$ and $G(y)$ by $F_n(x)$ and $G_n(y)$, the empirical distribution

functions of $\{X_i, i = 1, 2, \dots, n\}$ and $\{Y_i, i = 1, 2, \dots, n\}$, respectively.

- (3) Set $V_i^{x0*} = F_n^{-1}(U_i)$ and $V_i^{y0*} = G_n^{-1}(V_i)$, $i = 1, 2, \dots, n$.

This can be done as follows:

Obtain the order statistics $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$, then

$$V_i^{x0*} = \begin{cases} X_{(nU_i)} & \text{if } nU_i \text{ is an integer} \\ X_{(\lfloor nU_i \rfloor + 1)} & \text{if } nU_i \text{ is not an integer.} \end{cases}$$

The V_i^{y0*} can be obtained similarly.

- (4) Define $C_n(\alpha; \mathbf{Z}_n)$ by

$$P^*(\hat{\rho}_s(\mathbf{Z}_n^{0*}) \geq C_n(\alpha; \mathbf{Z}_n)) \cong \alpha, \quad (4)$$

where

$$\mathbf{Z}_n^{0*} = \{(V_1^{x0*}, V_1^{y0*}), (V_2^{x0*}, V_2^{y0*}), \dots, (V_n^{x0*}, V_n^{y0*})\},$$

and P^* denotes the conditional probability law of $\mathbf{Z}_n^{0*}$ given $\mathbf{Z}_n$.

We call this a semiparametric bootstrap test because a parametric family of copulas is assumed, but $F(x)$ and $G(y)$ are considered as unknown nuisance parameters.

In the next section we propose a completely nonparametric bootstrap test for ρ_s .

4. Nonparametric bootstrap test

Again, assume that $F(x)$ and $G(y)$ are unknown nuisance parameters. From

(1) it is clear that testing the hypothesis

$$H_0 : \rho_s(X, Y) = \rho_{s,0} \quad \text{vs.} \quad H_A : \rho_s(X, Y) > \rho_{s,0}$$

is equivalent to testing

$$H_0 : \rho_p(F(X), G(Y)) = \rho_{s,0} \quad \text{vs.} \quad H_A : \rho_p(F(X), G(Y)) > \rho_{s,0}.$$

As before, the test rejects H_0 if and only if

$$\hat{\rho}_s(\mathbf{Z}_n) \geq C_n(\alpha),$$

where

$$P_{H_0}(\hat{\rho}_s(\mathbf{Z}_n) \geq C_n(\alpha)) \cong \alpha.$$

In order to estimate the unknown critical value $C_n(\alpha)$ we proceed as follows:

Consider the following transformation of (X, Y)

$$\begin{aligned} X^0 &= \sqrt{3}(F(X) - 0.5) \left[\sqrt{\frac{1 + \rho_{s,0}}{1 + \rho_s(X, Y)}} + \sqrt{\frac{1 - \rho_{s,0}}{1 - \rho_s(X, Y)}} \right] + \\ &\quad \sqrt{3}(G(Y) - 0.5) \left[\sqrt{\frac{1 + \rho_{s,0}}{1 + \rho_s(X, Y)}} - \sqrt{\frac{1 - \rho_{s,0}}{1 - \rho_s(X, Y)}} \right] \end{aligned} \quad (5)$$

$$\begin{aligned} Y^0 &= \sqrt{3}(F(X) - 0.5) \left[\sqrt{\frac{1 + \rho_{s,0}}{1 + \rho_s(X, Y)}} - \sqrt{\frac{1 - \rho_{s,0}}{1 - \rho_s(X, Y)}} \right] + \\ &\quad \sqrt{3}(G(Y) - 0.5) \left[\sqrt{\frac{1 + \rho_{s,0}}{1 + \rho_s(X, Y)}} + \sqrt{\frac{1 - \rho_{s,0}}{1 - \rho_s(X, Y)}} \right]. \end{aligned} \quad (6)$$

Theorem 1 $\rho_p(X^0, Y^0) = \rho_{s,0}$.

Proof. It follows, after some tedious calculations, that $E(X^0 Y^0) = \rho_{s,0}$, $E(X^0) = E(Y^0) = 0$ and $E((X^0)^2) = E((Y^0)^2) = 1$. Hence,

$$\rho_p(X^0, Y^0) = \frac{Cov(X^0, Y^0)}{\sqrt{Var(X^0)}\sqrt{Var(Y^0)}} = \rho_{s,0}. \quad \blacksquare$$

Consider data $\mathbf{Z}_n^0 = \{(V_i^{x0}, V_i^{y0}), i = 1, 2, \dots, n\}$, where V_i^{x0} is given by the expression in (5) with:

- $F(X)$ replaced by $F_n(X_i) = \frac{R_i^x}{n}$ (if F is continuous), where $\{R_i^x, i = 1, 2, \dots, n\}$ denote the ranks of $\{X_i, i = 1, 2, \dots, n\}$.
- $G(Y)$ replaced by $G_n(Y_i) = \frac{R_i^y}{n}$ (if G is continuous), where $\{R_i^y, i = 1, 2, \dots, n\}$ denote the ranks of $\{Y_i, i = 1, 2, \dots, n\}$.

$1, 2, \dots, n$ denote the ranks of $\{Y_i, i = 1, 2, \dots, n\}$.

- $\rho_s(X, Y)$ replaced by $\rho_s(\mathbf{Z}_n)$.

V_i^{y0} is defined similarly using equation (6). Resampling now corresponds to sampling with replacement from the pairs $Z_n^0 = \{(V_i^{x0}, V_i^{y0}), i = 1, 2, \dots, n\}$. Refer to these resampled pairs as

$$\mathbf{Z}_n^{0*} = \{(V_i^{x0*}, V_i^{y0*}), i = 1, 2, \dots, n\}.$$

The bootstrap critical value $C_n(\alpha; \mathbf{Z}_n)$ is now chosen as in (4).

We can approximate $C_n(\alpha; \mathbf{Z}_n)$ by $C_{n,B}(\alpha; \mathbf{Z}_n)$ as follows:

- (1) Obtain a first bootstrap sample $\mathbf{Z}_n^{0*}$ and calculate $\hat{\rho}_p(\mathbf{Z}_n^{0*})$ (Pearson's correlation between $\{V_i^{x0*}, i = 1, 2, \dots, n\}$ and $\{V_i^{y0*}, i = 1, 2, \dots, n\}$). Denote this by T_1^* . It is important to note that one needs to calculate $\hat{\rho}_p(\mathbf{Z}_n^{0*})$ and not $\hat{\rho}_s(\mathbf{Z}_n^{0*})$.
- (2) Independently repeat step (1) a number B times to obtain B bootstrap replications $T_1^*, T_2^*, \dots, T_B^*$.
- (3) Obtain the order statistics $T_{(1)}^* \leq T_{(2)}^* \leq \dots \leq T_{(B)}^*$.
- (4) Set $C_{n,B}(\alpha; \mathbf{Z}_n) = T_{(\lfloor B(1-\alpha) \rfloor)}^*$, where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x .

Hence, reject the null hypothesis if and only if

$$\hat{\rho}_s(\mathbf{Z}_n) \geq \hat{C}_n(\alpha; \mathbf{Z}_n).$$

5. Monte-Carlo study

In this section we present the results of a Monte-Carlo study where we compare the estimated sizes and estimated powers of the semiparametric test (denoted by (S)), the nonparametric test (denoted by (N)) and the asymptotic normal test defined in (3) (denoted by (Norm)) for sample sizes $n = 20, 30, 50$ and 100 , respectively. *Estimated powers are only presented in cases where the estimated sizes of the tests were comparable.* A nominal significance level of $\alpha = 0.05$ was used. Bootstrap critical values were based on $B = 1000$ independent

replications. All calculations were done using double precision arithmetic in FORTRAN and routines from the IMSL library. All size (or power) estimates were calculated as the proportion of 20 000 independent Monte-Carlo samples that resulted in H_0 (or H_A) being rejected, using the associated bootstrap critical values. All standard errors of the estimated probabilities are less than or equal to $\sqrt{0.25/20\,000} = 0.0035$. Data were generated from the

- Cuadras-Augé copula given by $C_\theta(u, v) = [\min(u, v)]^\theta [uv]^{1-\theta}$, $0 \leq \theta \leq 1$, and using (2) we have that $\rho_s = \frac{3\theta}{4-\theta}$;

- Plackett copula given by $C_\theta(u, v)$

$$= \frac{1 + (\theta - 1)(u + v) - \sqrt{[1 + (\theta - 1)(u + v)]^2 - 4uv\theta(\theta - 1)}}{2(\theta - 1)}, \quad \theta >$$

$$0, \theta \neq 1, \text{ and from (2) it follows that } \rho_s = \frac{\theta + 1}{\theta - 1} - \frac{2\theta \ln \theta}{(\theta - 1)^2};$$

- Bivariate normal distribution for which (2) yields $\rho_s = \frac{6}{\pi} \arcsin\left(\frac{\theta}{2}\right)$;
- Raftery's bivariate exponential distribution. The Raftery distribution is generated by considering

$$X = (1 - \theta)Z_1 + IZ_3 \quad \text{and}$$

$$Y = (1 - \theta)Z_2 + IZ_3,$$

where Z_1, Z_2 and Z_3 are independent and identically distributed standard exponential random variables and I is a Bernoulli random variable with parameter θ , independent of Z_i , $i = 1, 2, 3$. This bivariate exponential distribution is due to Raftery (1984). In this case, using (2) once again, we find that $\rho_s = \frac{\theta(4 - 3\theta)}{(2 - \theta)^2}$.

For the semiparametric test we used $F(x) = 1 - e^{-x}$ and $G(y) = 1 - e^{-y}$ as the marginal distributions. The semiparametric test was only evaluated for the data generated from the two copulas. In Tables 2 and 4, $\rho_{s,A}$ denotes the value of ρ_s under H_A .

Table 1. Estimated size for testing Spearman's rho

Copula	$\rho_{s,0}$	n	(S)	(N)	(Norm)
Cuadras-Au�e	$\rho_{s,0} = 0.1$	20	0.052	0.045	0.064
		30	0.050	0.045	0.060
		50	0.054	0.048	0.059
		100	0.054	0.051	0.058
	$\rho_{s,0} = 0.5$	20	0.051	0.042	0.114
		30	0.050	0.042	0.105
		50	0.052	0.041	0.098
		100	0.048	0.042	0.092
Plackett	$\rho_{s,0} = 0.1$	20	0.050	0.046	0.050
		30	0.051	0.047	0.050
		50	0.050	0.047	0.049
		100	0.053	0.050	0.052
	$\rho_{s,0} = 0.5$	20	0.050	0.056	0.067
		30	0.051	0.054	0.062
		50	0.051	0.053	0.064
		100	0.051	0.047	0.063

Table 2. Estimated power for testing Spearman's rho

Copula	$\rho_{s,A}$	n	(S)	(N)	(Norm)
Plackett	$\rho_{s,A} = 0.2$	20	0.108	0.103	0.106
		30	0.132	0.129	0.130
		50	0.171	0.169	0.171
		100	0.267	0.263	0.266
	$\rho_{s,A} = 0.3$	20	0.218	0.219	0.215
		30	0.284	0.271	0.282
		50	0.407	0.395	0.402
		100	0.654	0.648	0.652

Table 3. Estimated size for testing Spearman's rho

Distribution	$\rho_{s,0}$	n	(N)	(Norm)
Bivariate normal	$\rho_{s,0} = 0.1$	20	0.048	0.051
		30	0.048	0.052
		50	0.045	0.047
		100	0.050	0.050
	$\rho_{s,0} = 0.5$	20	0.051	0.053
		30	0.051	0.054
		50	0.049	0.051
		100	0.048	0.053
Bivariate exponential	$\rho_{s,0} = 0.1$	20	0.046	0.060
		30	0.046	0.058
		50	0.047	0.053
		100	0.047	0.052
	$\rho_{s,0} = 0.5$	20	0.056	0.077
		30	0.054	0.075
		50	0.054	0.076
		100	0.057	0.076

Table 4. Estimated power for testing Spearman's rho

Distribution	$\rho_{s,A}$	n	(N)	(Norm)
Bivariate normal	$\rho_{s,A} = 0.6$	20	0.141	0.143
		30	0.178	0.176
		50	0.243	0.241
		100	0.389	0.391
	$\rho_{s,A} = 0.7$	20	0.326	0.332
		30	0.458	0.470
		50	0.651	0.665
		100	0.908	0.912

From Tables 1–4 we can conclude the following:

- (1) All estimated sizes of (S) are very close to the nominal significance level, as one would expect.
- (2) (Norm) performs satisfactorily (in terms of estimated sizes) when the underlying copulas and bivariate distributions have a weak Spearman's correlation ($\rho_s = 0.1$). However, if the underlying distributions have a

stronger Spearman's correlation ($\rho_s = 0.5$), (Norm) tends to be "liberal". See, e.g., the results obtained when using the Cuadras-Augé copula.

- (3) The estimated sizes of (N) are close to the nominal significance level, even if the underlying distribution has a relatively strong Spearman's correlation ($\rho_s = 0.5$).
- (4) The performances of the three tests are very similar with regard to estimated powers (see Tables 2 and 4).

Based on the overall good performance with regard to estimated size and power, we recommend that the nonparametric test (N) be used for testing the hypothesis

$$H_0 : \rho_s = \rho_{s,0} \quad \text{vs.} \quad H_A : \rho_s > \rho_{s,0}.$$

Naturally, if one knows the underlying copula, the semiparametric test should be the best choice for testing this hypothesis.

6. Final remarks

In this paper bootstrap-based tests were developed for Spearman's rho that performed satisfactorily. For the construction of the nonparametric test it was necessary to develop appropriate transformations such as those given in (5) and (6) in order to "mimic" H_0 in the bootstrap world. Currently we are also studying the problem of developing semiparametric and nonparametric bootstrap tests for other measures of association.

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